

Technology Stack

Date	30 September 2026
Team ID	SWTID-2026-6449
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	2 Marks

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, Tailwind CSS, TypeScript	Enables modular component lifecycle management for fluid split-screen template generation; type safety via TypeScript limits structural form bugs (Deepan).
2	Backend / Server-Side	Python, FastAPI	Asynchronous capabilities ensure lightweight, ultra-high-speed routing between user endpoints and underlying neural networks (M Mohammed Abdul Rahman / Lingaraj S).
3	Database / Data Storage	PostgreSQL, Redis	PostgreSQL ensures strict ACID compliance for sensitive legal templates and multi-party signature records, while Redis acts as a low-latency user session cache.
4	Cloud / Hosting / Deployment	Docker, AWS (Amazon ECS & EKS)	Dockerizes runtime microservices across matching development environments, scaling resources automatically during high traffic enterprise processing windows.
5	Version Control & CI/CD	GitHub, GitHub Actions	Maintains absolute code transparency between the 4 team members with automated linting, security checking, and deployment pipelines.
6	Third-Party APIs / Other Tools	OpenAI API (GPT-4o/o1), DocuSign API	Powers the core semantic translation of plain speech into legal boilerplate text alongside verified multi-party identity execution signing workflows (Sakthi).

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